

TrES-5b

R-0910

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240812 · created 2026-07-29T06:06:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460534.7768 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.049
Transit depth (Rp/Rs)^2	1.58 +/- 1.22 [%]
Orbital Inclination (inc)	89.37 +/- 2.87
Airmass coefficient 1 (a1)	1.07 +/- 0.024
Airmass coefficient 2 (a2)	-0.06 +/- 0.23
Scatter in the residuals of the lightcurve fit is	2.29 %
Best Comparison Star	#2 - [372, 353]
Optimal Method	PSF photometry
Transit Duration (day)	0.0828 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.126 ± 0.049 vs 0.142 (deviation 0.33σ)
Duration fit vs published	0.0828 d vs 0.0758 d (deviation 1.06σ)
Mid-time O-C	9.4 min
Depth SNR	2.6
Residual scatter	2.29 %

Light curve

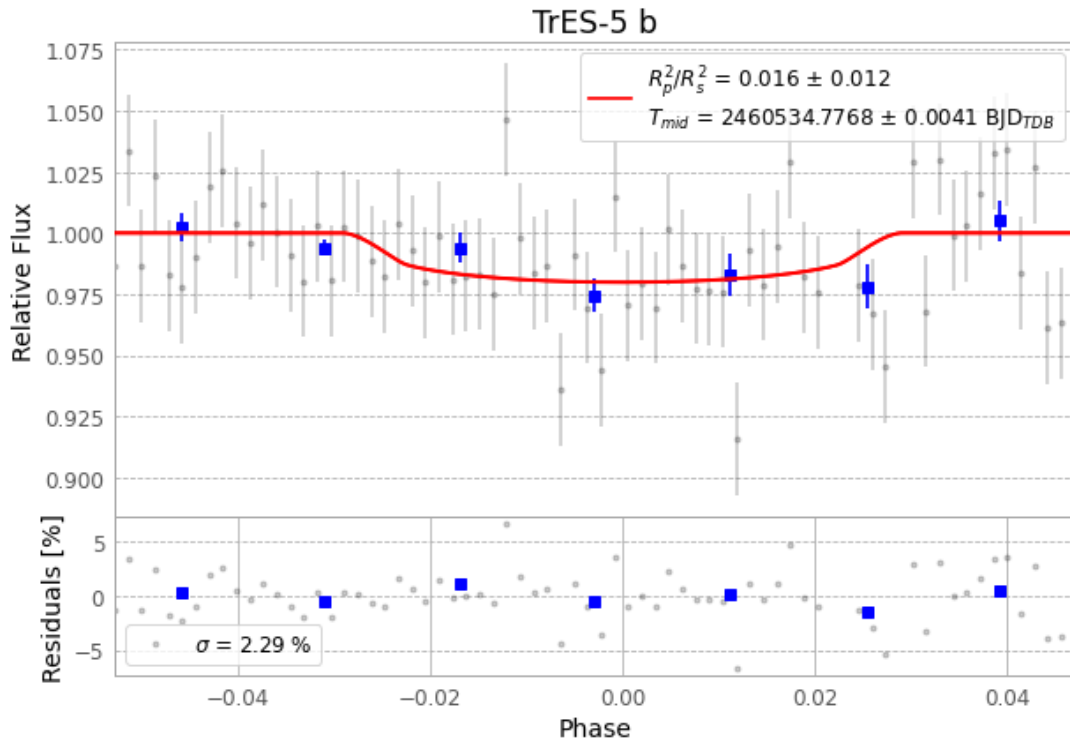


Figure 1 — FinalLightCurve_TrES-5 b_2024-08-11.png (SHA-256 12a2531bf3da6e56...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32efc26da2176dd3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5240812044219.FITS → TRES-5240812081515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240812.log (24118c9dc25a...), FinalLightCurve_TrES-5 b_2024-08-11.png (12a2531bf3da...), FinalLightCurve_TrES-5 b_2024-08-11.csv (7d66c94e348f...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 06:06Z.